PSYCHOLOGY

Chapter 1:
Research Methods

Welcome to AnithUncommon's **Sample Resources!**

We are assuming you are reading this as a student. So what will **YOU** expect from this sample?

In Psychology, Research Methods, 1.1-1.11…

This sample will be divided into **three** parts: Introduction to Samples (this part), Lesson Objectives (What is Expected From Students), and most crucially, Syllabus Contents.

Note that the 'Syllabus Content' does NOT stand as a complete syllabus itself; rather, it is a tight compilation of notes gathered by Sharia Shifa to create a comprehensive sample for students.

If you want a complete version of these notes and think that you need help in Psychology, join our nonprofit!

Gmail: anithuncommon
Instagram handle: @anithuncommon

This sample is made thanks to, Sharia Shifa

— Made by students, for students.

Lesson Objectives:

At the end of the lesson, students should be *able* to…

- **State** the strengths and weaknesses of various experimental and non-experimental methods.
- **Understand** the types of variables.
- **Compare** research based on reliability, validity, and ethical standards.
- **Analyze** and **interpret** quantitative and qualitative data using statistical measures.

Reflective Questions to Consider:

Psychology aims to study human behavior scientifically, but human behavior is complex and difficult to control. Different research methods offer different strengths, yet each comes with its own limitations.

So, think about these:

- Why do psychologists use experiments instead of just observing people?
- What is the difference between an independent variable (IV) and a dependent variable (DV)?
- Which method do you think is more reliable: experiments, observations, or questionnaires? Why?
- Why is it important to follow ethical rules when studying people?

Unit 1: Research Methods

Tip!

- You might get confused between different research methods and variables at first. Try to focus on what each method is used for and how the independent and dependent variables are connected. Once you understand the purpose behind each method, everything becomes easier to remember.

Activities in Class:

- Lesson Breakdown
- Scenario Activity: Identify IV (independent variable) and DV (dependent variable) from different scenarios.
- Ethics Court: Debate real-world case studies to decide if they meet modern ethical standards.

1.1 Experiments and Variables

A. Types of Experiments

a. **Laboratory Experiment:** A laboratory experiment is carried out in a highly controlled setting, where the researcher can carefully manage all variables. This high level of control makes it easier to establish cause-and-effect relationships. However, because the environment is artificial, participants may not behave as they normally would in real life.

- ✔ Conducted in a *highly controlled environment* where variables can be carefully managed.
- ✔ This allows high control over results, but behavior may feel *artificial*.

b. **Field Experiment:** A field experiment takes place in a natural environment, such as a school or public place. This allows behaviour to be more realistic and natural. However, since the researcher has less control over outside factors, it becomes harder to ensure that only the intended variable is affecting the results.

- ✔ Conducted in a *natural environment* where

Mentor's Notes

To understand research methods, you must identify the patterns between different methods rather than memorizing them individually. Each method has a specific purpose: experiments are used to find cause-and-effect relationships, observations are used to study natural behavior, and self-reports are used to understand thoughts and feelings. Notice how these

participants behave more realistically. ✔ However, there is *less control over external factors*. c. **Natural Experiment:** In a natural experiment, the independent variable is not controlled by the researcher and instead occurs naturally. This method is useful when it would be unethical or impossible to manipulate variables. However, because of the lack of control, it is more difficult to confidently establish cause-and-effect. ✔ The independent variable *occurs naturally* and is NOT controlled by the researcher. ✔ Useful when *variables cannot be ethically manipulated*, but cause-and-effect is harder to establish. **NOTE:** Do not confuse the field experiment with the natural experiment. Field/ true experiment: A researcher may want to know which method would lead to better learning among students' lecture or demonstration method. For this, a researcher may prefer to conduct an experiment in the school. The researcher may select two groups of participants; teach one group by demonstration method and another group by the normal teaching method for sometime. Then they may compare their performance at the end of the learning session. In such types of experiments, the control over relevant variables is less than what we find in laboratory experiments. Natural/Quasi experiment: If you want to study the effect of an earthquake on children who lost their parents, you cannot create this condition artificially in the laboratory. In such situations, the researcher adopts the method of quasi (the Latin word meaning "as if ") experimentation. In the experimental group we can have children who lost their parents in the earthquake and in the control group children who experienced the earthquake but did not lose	**methods differ in terms of control, realism, and type of data collected.** **Why does this matter? Because psychology is not just about collecting information, but about choosing the right method to answer a question. For example, if you want control, you use experiments. If you want realistic behavior, you use observations. If you want opinions or personal experiences, you use self-reports. Understanding this connection helps you evaluate which method is most appropriate in different situations.** **(I have added a table at the end which can help you understand this better)** **Also note that variables (IV, DV, EV) are present in**

their parents. Thus, a quasi experiment attempts to manipulate an independent variable in a natural setting using naturally occurring groups to form experimental and control groups.

Feature	Field Experiment	Natural experiment
IV manipulation	Yes, by the researcher	No, occurs naturally
Control	Moderate	Negligible
Setting	Real-world	Real-world

B. Variables in Research

a. Independent Variable (IV): The independent variable is the factor that the researcher changes in order to observe its effect on behaviour. It is the "cause" in an experiment.

- ✔ The variable that is changed or manipulated by the researcher.

b. Dependent Variable (DV): The dependent variable is what the researcher measures. It shows the effect of the independent variable and is therefore the "outcome" of the study.

- ✔ The variable that is measured to observe the effect of the IV.

c. Extraneous Variables (EV): Extraneous variables are other factors that could influence the results if they are not controlled. These variables can reduce the accuracy of the study.

- ✔ Other factors that may affect the results if not controlled.

NOTE: EV is like that scary teacher who hates you for no reason. So you should try to avoid them so it doesn't ruin your day– the experiment.

almost every study. These variables determine how reliable and valid your results are. If variables are not controlled properly, the entire study can become inaccurate. Therefore, always link methods with variables and think about how they affect the outcome of the research.

Tip!

Try to connect each research method to a real-life example. For instance, use experiments to test effects (like sleep and memory), observations to study behavior (like crowd reactions), and questionnaires to understand opinions.

Example:
If studying how sleep affects memory:

IV → amount of sleep (what is changed)
DV → memory test score (what is measured)
EV → temperate of the room, comfort, etc

1.2 Self-Reports

A. Questionnaires: A questionnaire is a set of written questions used to collect data from participants. These can be open-ended, allowing detailed answers, or closed-ended, where responses are limited to specific options. Questionnaires are useful for collecting large amounts of data quickly, but responses may not always be honest or accurate.

- ✔ A set of written questions used to gather data from participants.
- ✔ Can be open-ended (detailed answers) or closed-ended (fixed responses).

B. Interviews: An interview involves direct interaction between the researcher and the participant, allowing for more detailed responses. Interviews can be structured, with fixed questions, or unstructured, where the conversation is more flexible. Although interviews provide deeper insights, they can sometimes be influenced by interviewer bias or social desirability.

- ✔ Direct interaction with participants to collect deeper responses.

Can be:
- Structured (fixed questions)
- Unstructured (flexible discussion)

Strength: Rich, detailed data
Weakness: Can be biased or dishonest

1.3 Case Studies

A case study is a detailed investigation of a single person or a very small group, usually over a long period of time. Instead of collecting quick data from many people, psychologists focus deeply on one case to understand complex or rare behaviour**.**

Example: if someone has a rare memory disorder, studying them closely can reveal how memory works in ways large experiments cannot.

- ✔ In-depth investigation of a single person or small group.
- ✔ Used to explore rare or complex behaviors.

Strength: Detailed insight
Weakness: Cannot be generalized to everyone

1.4 Observations

Observations involve watching and recording behavior as it naturally happens, rather than asking questions or manipulating variables.

<u>Types of Observation:</u>

Structured → The researcher uses a checklist of specific behaviors to look for, making the data more organized and easier to compare.
- Pre-planned behavior checklist

Unstructured → The researcher observes freely without a fixed plan, allowing unexpected behaviors to be recorded.
- Free observation

Participant → The researcher becomes part of the group being studied, which may give deeper insight but can affect objectivity.
- Researcher joins the group

Non-participant → The researcher observes from a

distance without getting involved, reducing influence on behavior. • Researcher observes from outside Strength: Captures real-life behavior in natural settings. Weakness: Results may be affected by observer bias (the researcher interpreting behavior subjectively). **1.5 Correlations** Correlation measures the relationship between two variables to see if they are connected in some way. ▪ Measures the relationship between two variables. Positive correlation → both increase together (e.g., more study time → higher test scores) Negative correlation → one increases, other decreases (e.g., more stress → lower performance) **Important:** Correlation does **NOT** mean causation. Just because two things are related does not mean one causes the other. → *Causation* means one variable directly causes a change in another (cause → effect). Example: Ice cream sales and drowning rates both increase in summer — but ice cream does NOT cause drowning. **1.6 Research Process** Psychological research follows a step-by-step process to ensure results are logical and reliable. Steps followed in conducting research: • Form a **hypothesis** (a testable prediction) • Choose a **method** (experiment, observation, etc.)	

- **Collect data** from participants
- **Analyze results** using statistics or patterns
- Draw a **conclusion** based on findings

→ This process ensures research is systematic and not based on random guesses.

1.7 Variables

Variables are the foundation of any research study, and how they are handled affects the accuracy of results.

• **Defined clearly (operationalized)**
→ Variables must be explained in a measurable way (e.g., "happiness" = score on a questionnaire)

• **Controlled carefully**
→ Other influencing factors must be minimized to avoid affecting results

• **Measured accurately**
→ Reliable tools must be used to ensure consistent results

Poor handling of variables can make the entire study unreliable.

1.8 Sampling

Sampling is the process of selecting participants for a study. Since studying everyone is impossible, psychologists choose a smaller group to represent the population.

Types:

• **Random sampling**
→ Everyone has an equal chance of being selected (reduces bias)

• **Opportunity sampling**

→ Participants are chosen based on availability (quick but biased) • **Volunteer sampling** → Participants choose to take part (may attract certain personality types) The type of sampling used affects how fair and reliable the results are. **1.9 Data and Analysis** Two types of data: • **Quantitative data** Numerical data (e.g., scores, percentages) → easy to analyze • **Qualitative data** Descriptive data (e.g., opinions, feelings) → more detailed but harder to measure Data is analyzed using: • Averages (mean, median, mode) • Graphs and patterns **1.10 Ethics** Ethics are the rules psychologists must follow to protect participants during research. Since studies often involve real people, it is important to ensure their safety, privacy, and dignity are respected at all times. **Key Principles:** • **Informed consent**: Participants must be fully aware of what the study involves before agreeing to take part. This means they understand the purpose, procedure, and	

any potential risks. • **Confidentiality**: Personal information must be kept private. Names and identities should not be revealed in the results. • **Right to withdraw**: Participants can leave the study at any time without giving a reason, and without any negative consequences. • **Protection from harm**: Participants should not be exposed to physical or psychological harm. The study must not cause stress, anxiety, or discomfort beyond normal levels. These guidelines ensure that research is conducted responsibly and respectfully. **1.11 Evaluating Research** After a study is conducted, psychologists must evaluate how good and trustworthy the research is. This helps determine whether the findings can be accepted or questioned. • **Reliability (Consistency)** → Refers to whether the results can be repeated. → If the same study is done again and gives similar results, it is considered reliable. • **Validity (Accuracy)** → Refers to whether the study actually measures what it claims to measure. → A study can be reliable but still not valid if it measures the wrong thing. • **Bias (Fairness)** → Occurs when results are influenced by personal opinions, expectations, or unfair sampling.	

→ Bias reduces the credibility of the research.

→ A strong study should be reliable, valid, and as free from bias as possible.

Checking quality of research using:

- Reliability → consistency
- Validity → accuracy
- Bias → fairness

Method	**Purpose**	**Strengths**	**Weaknesses**
Experiment	Find cause and effect	High control, clear results	Not realistic (lab setting)
Observation	Study natural behavior	Realistic data	Observer bias, low control
Self-report	Understand thoughts/feelings	Quick and easy data collection	Dishonest or biased answers
Case Study	Study rare/complex behavior	Very detailed data	Cannot generalize
Correlation	Find relationships between variables	Shows patterns/trends	Cannot prove cause and effect

Mentors’ Sources:

PSYCH book full.pdf

Chapter 2 - Methods of Enquiry in Psychology.PMD

British Psychological Society (BPS) Code of Ethics and Conduct

American Psychological Association (APA) Ethical Guidelines

CASE STUDY- Patient H.M. (Henry Molaison)